BRIDGET NAYEBARE

nayebarebridget2017@gmail.com

nayebarebridget97@gmail.com

FINAL PROJECT PRESENTATION.

FROM SEEDS TO SUCCESS: THE GREEN ENGINE.

Can Agriculture Predict National Wealth?

The answer is YES.

When we think of a wealthy country, we often think of tech hubs, big banks or even huge factories. But for much of the world, the real engine of wealth starts in the soil. Farming isn't just about food, it's about money.
I'm going to show you how I used Artificial Intelligence to prove that a nation's oldest industry (Agriculture) is actually one of the most powerful predictors of its economic future.

GOAL: To use Deep Learning to understand the relationship between a country's agricultural output and its overall Gross Domestic Product (GDP).

DATA SOURCE:

We shall use;
FAO Macro-Economic Indicators dataset

Provided via World Bank Data360

From Food and Agriculture Organization

Link: https://data360files.worldbank.org/data360-data/data/FAO_MK/FAO_MK.csv

DATA DESCRIPTION.

The dataset contains various economic agricultural indicators related to Gross Domestic Product (GDP) and agricultural value added across different countries and time periods.

It has **72,198 rows and 37 columns** after data cleaning.

Key columns observed from rows and columns include:

REF AREA LABEL: This column represents the country or geographical area for which the data is recorded.

INDICATOR LABEL: Describes the economic indicator being measured (e.g., 'Gross Domestic Product', 'Value Added (Agriculture)').

TIME PERIOD: Represents the year for which the data is recorded, ranging from **1970 to 2024**.

OBS VALUE: The observed numerical value for the given indicator in the specified time period and region.

UNIT MULT: Indicates the multiplier for the OBS Values (e.g., 'Millions').

Also noted the following about numerical columns:

TIME PERIOD: Ranges from 1970 to 2024, with a mean around 1999, indicating a distribution across several decades.

OBS VALUE: Shows a wide range, from 0 to 1.98e+11, suggesting significant variability in economic values across indicators and countries.

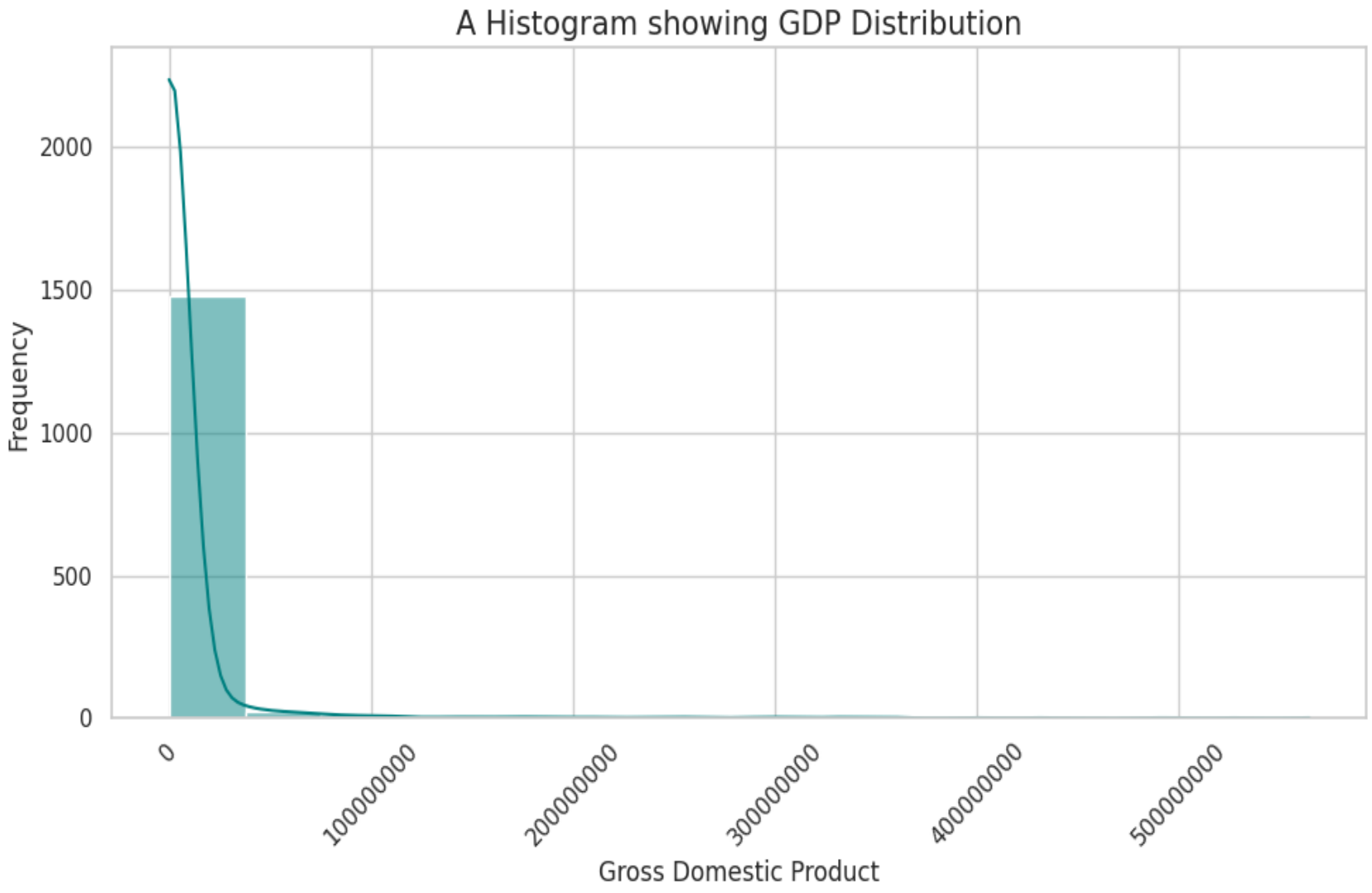
The data also includes categorical features like Age, Sex and Urbanization which are not used in the current model but could provide further insights into specific demographics.

UVIVARIATE ANALYSIS

HISTOGRAM

OBSERVATIONS:

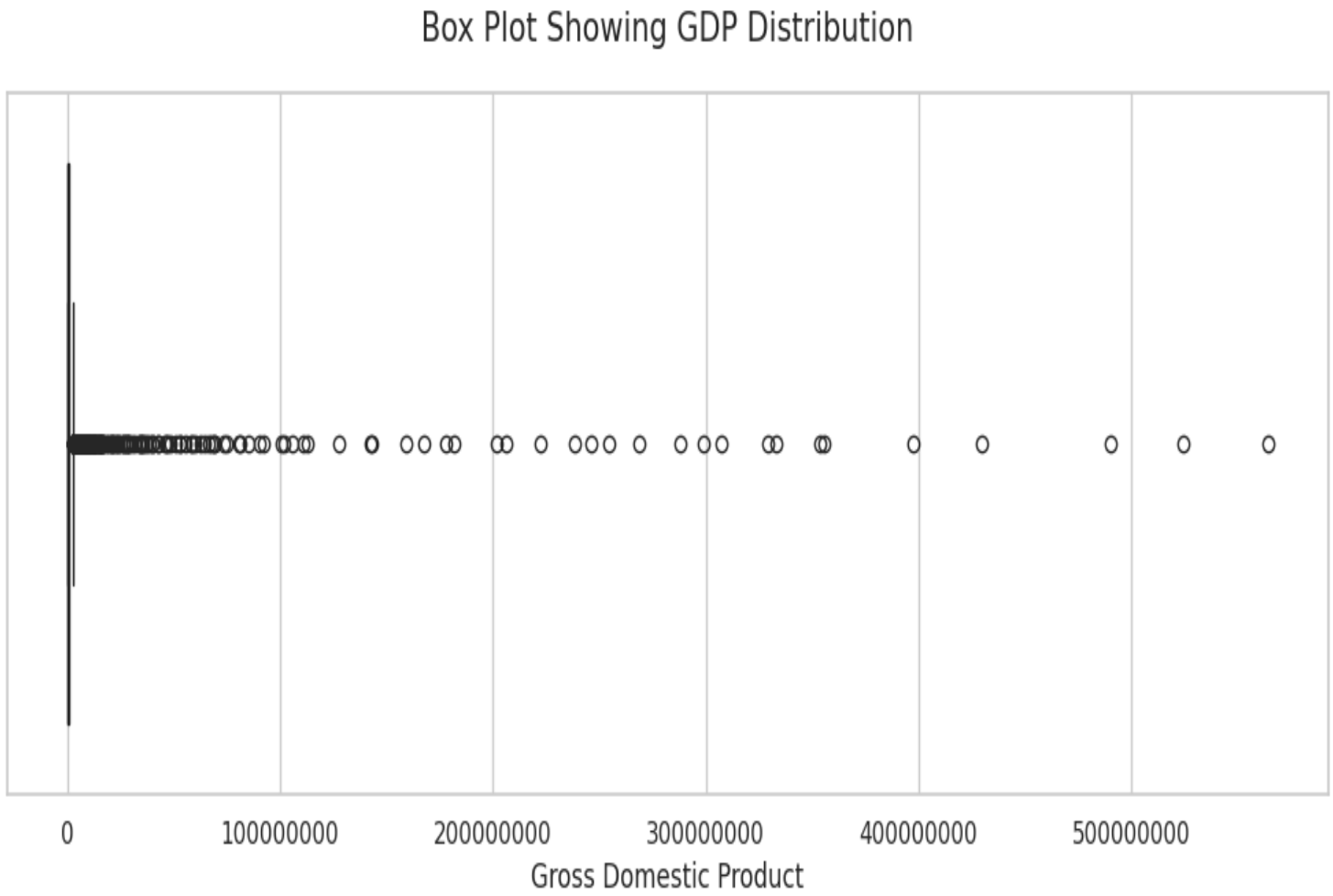
- ✓ The histogram shows the distribution of the GDP which is highly rightly skewed.
- ✓ A large concentration of the data points is found at the lower end of the GDP spectrum which indicates that majority of the countries in the data set have relatively lower GDP Values.
- ✓ The tail of the distribution extends significantly to the right, , suggesting the presence of a few countries with exceptionally high GDPs.



BOX PLOT

OBSERVATIONS:

- The box plot visually reinforces the findings from the histogram, confirming the highly skewed distribution of GDP.
- The box, representing the interquartile range (IQR), is very small and positioned at the lower end of the scale, indicating that the majority of GDP values are clustered within a narrow, lower range.
- There are numerous individual points extending far beyond the upper whisker, which are clearly identifiable as outliers. These outliers represent countries with significantly higher GDP values compared to the median.
- The extreme length of the upper whisker and the presence of many outliers underscore the vast economic disparities among the countries in the dataset.

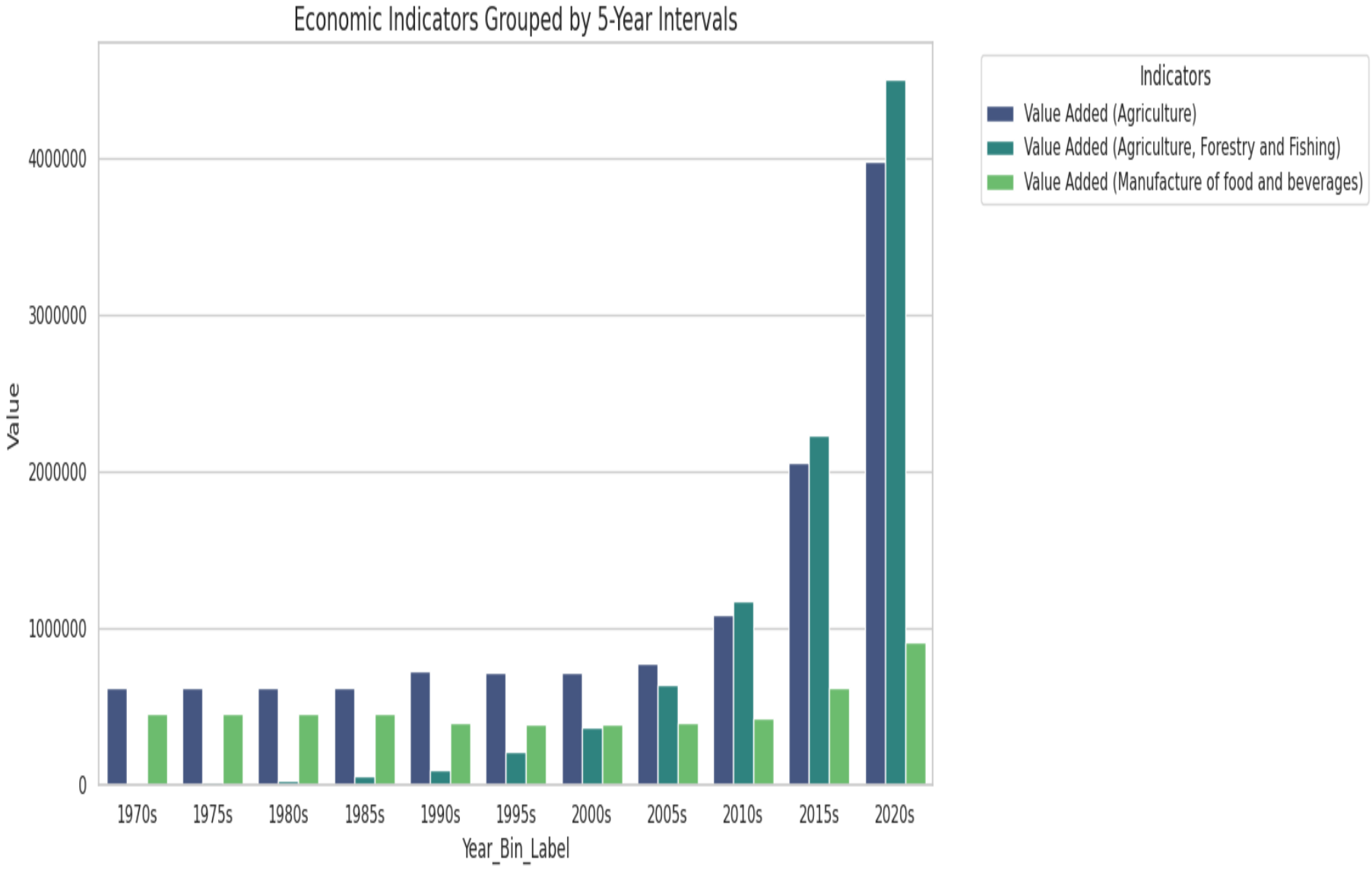


MULTVARIATE ANALYSIS

GROUP BAR GRAPH

OBSERVATIONS:

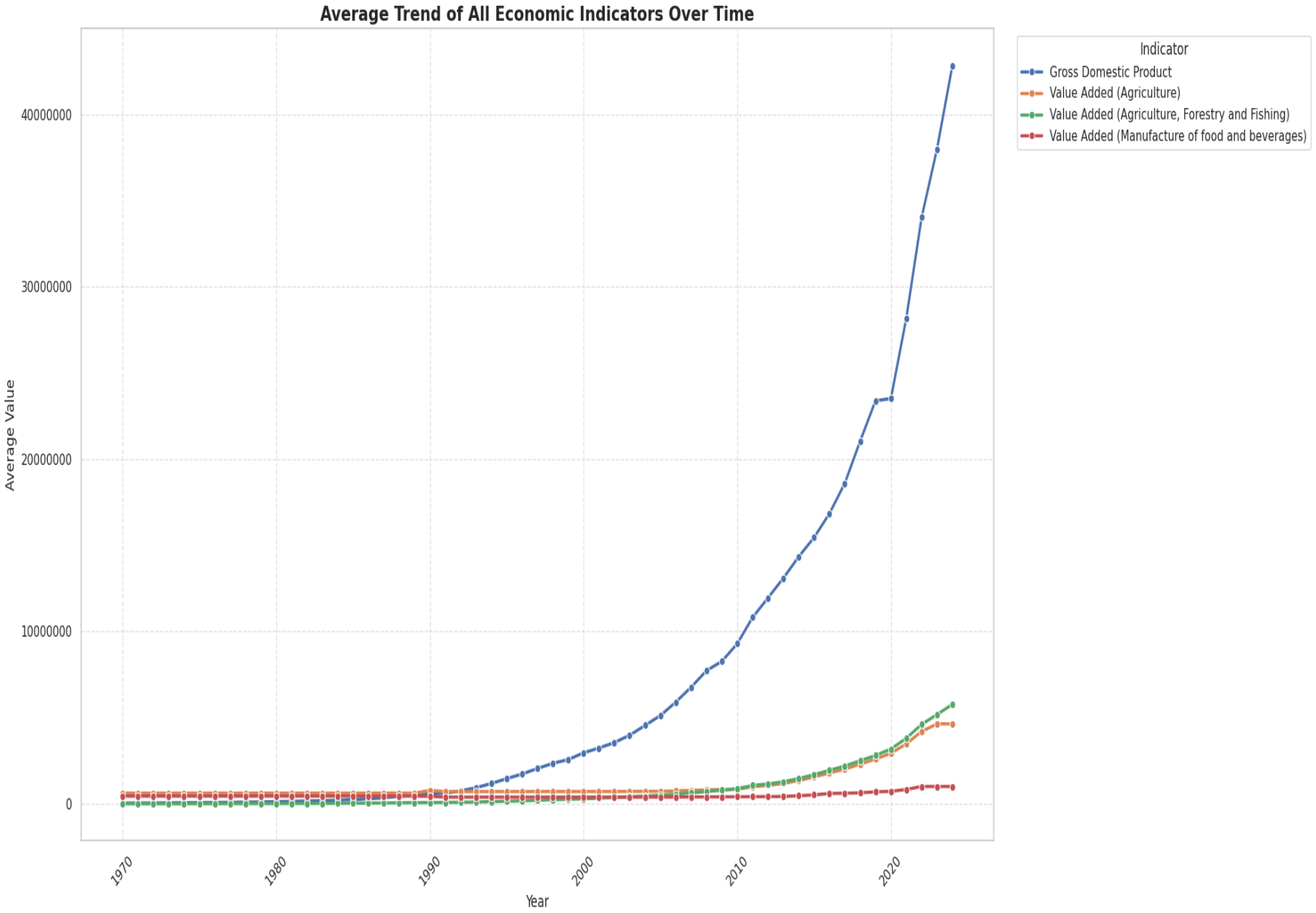
- This grouped bar graph visualizes the average trends of various economic indicators across 5-year intervals, providing insights into long-term economic shifts.
- There is a clear and consistent **upward trend** in the mean values for all indicators over the decades. This suggests sustained economic growth and development with in the sectors presented.
- Value Added (Agriculture) and Value Added (Agriculture ,Forestry and Fishing) generally exhibit higher average values compared to Value Added (Manufacture of food and beverages) across most periods especially in earlier decades. This highlights the role of primary sectors in economies as covered by this dataset.
- Value Added(Manufacture of food and beverages) becomes more noticeable in the later periods indicating increased industrialization and diversification of economies over time.



LINE GRAPH

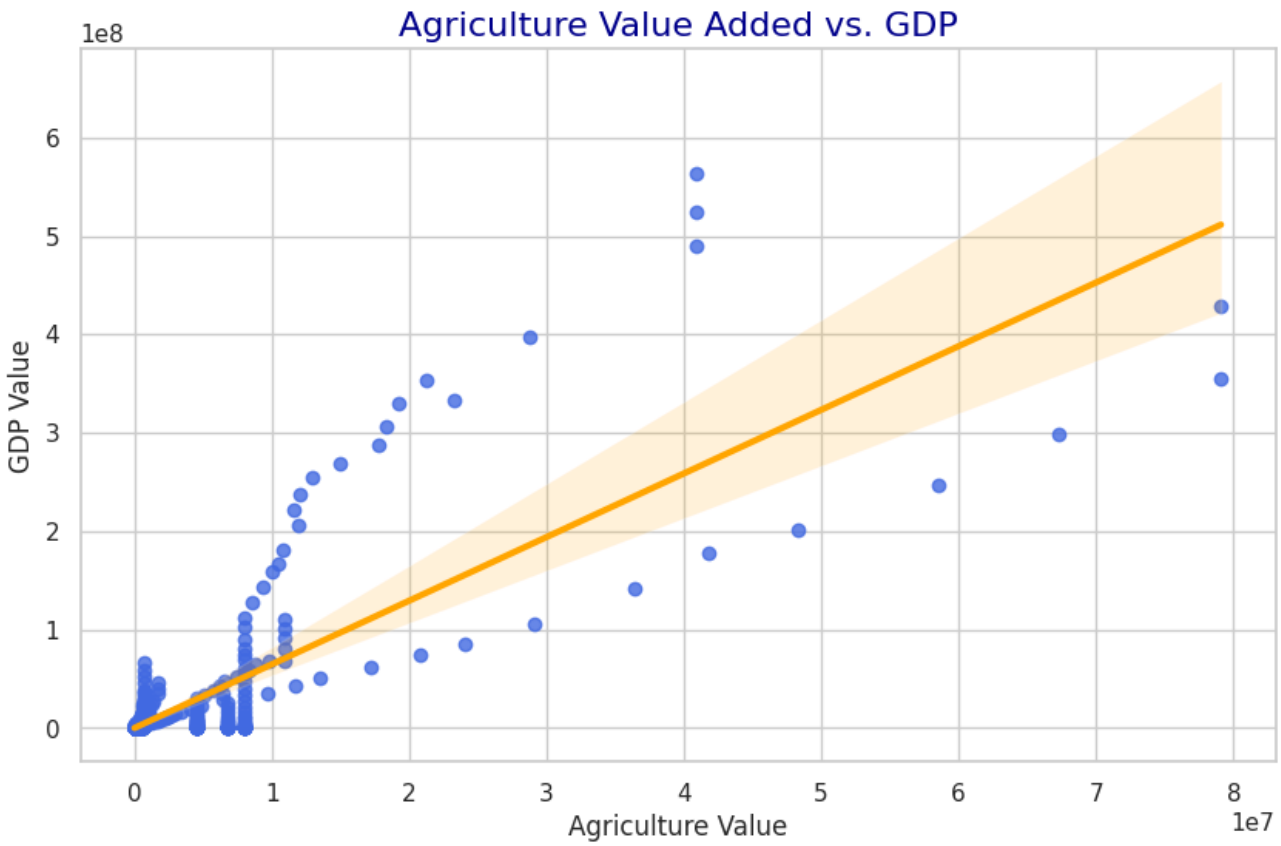
OBSERVATIONS;

- The line plot illustrates the average trend of all key economic indicators over the entire time period (1970-2024).
- All indicators show a **strong and consistent upward trend** across the years, signifying overall economic growth and development in the countries represented.
- Gross Domestic Product demonstrates the most substantial growth, with its average value increasing significantly over the decades. This indicates a general expansion of overall economic output.
- The Value Added (Agriculture) and The Value Added(Agriculture, Forestry and Fishing) lines generally track closely and show a steady increase, often maintaining higher average values than Value Added (Manufacture of food and beverages)
- Value Added (Manufacture of food and beverages) also shows an increasing trend, although its absolute average values remain lower than the other indicators for most of the period.



SCATTER PLOT

- The scatter plot visualizes the relationship between Value Added (Agriculture) and Gross Domestic Product with a regression line indicating the general trend.
- There is a clear **positive correlation** between Value Added (Agriculture) and Gross Domestic Product. As the value added by the agriculture sector increases, the overall GDP tends to increase as well.
- The spread of the data points, particularly at higher values, suggests that while there's a general positive trend, the relationship is not perfectly linear and can vary across different economic contexts or development stages.
- The majority of data points are concentrated at the lower end for both variables, which is consistent with the skewed distributions observed in earlier univariate analyses. This shows that for most instances, both agricultural output and overall GDP are relatively low, with fewer instances of very high values.
- The positive correlation between Value Added (Agriculture) and Gross Domestic Product is an indicator that Value Added (Agriculture) is a significant predictor of Gross Domestic Product



DEEP LEARNING

This where a neural network teaches a machine to process data in the same way a human brain would. Our data set is a time series data set and with numeric values, we decided to use a regression Model (Random Forest) so as to have a deeper understanding of the relationship between different features and the target.

This requires us to do feature Engineering,

What is feature engineering?

It's the process of creating new, more informative "clues" for our models from the data we already have. Instead of just using the raw numbers, we transform and combine them to reveal deeper patterns and relationships that help our models make more accurate predictions.

To ensure the Deep Learning model (and subsequent steps) receives the correctly engineered features, we re-run the feature engineering process here. This creates a Data Frame with all the required lagged and rolling mean features for the selected indicators. We then define the X (features) and Y (Target).

Why Feature Engineering?

Imagine trying to predict tomorrow's weather just by knowing today's temperature. It's helpful, but knowing yesterday's temperature, the average temperature over the last week and how much it rained recently would give you a much better picture!

This is exactly what we are trying to do.

In our GDP prediction project, this is crucial because:

1. **Capturing Time-Based Patterns:** Economic data changes over time. Just like a country's GDP isn't independent of its past, our models need to understand historical context.
2. **Lagged Features:** We create features that represent an indicator's value from the *previous year*. This tells the model, "What was the agricultural output last year?" which is often a strong hint for this year's economic activity.
3. **Rolling Statistics:** We calculate average values over a period (e.g., the last 3 years). This helps the model identify longer-term trends, smoothing out any year-to-year fluctuations and giving a sense of sustained growth or decline.
4. **Improving Model Intelligence:** By providing these richer, more insightful features, we empower our models to learn more effectively. They can connect the dots between current economic activity and its historical momentum, leading to more robust and reliable GDP forecasts.

- By carefully engineering these features, we give our models a more complete story of the economic landscape, significantly boosting their predictive power.

Preparing Our Data for the Model: Training and Scaling

Before our machine learning models (like the Random Forest or Neural Network) can learn, we need to get our data ready. This involves a few critical steps:

Separating Clues from Answers (Features and Target):

- We first identify what we want to predict: the **Gross Domestic Product (GDP)**. This is our "answer" or **target variable**.
- All the other economic indicators, such as "Value Added (Agriculture)" and its lagged and rolling mean versions, are our "clues" or **features**. These are what the model will use to make its predictions.

Smoothing Out the "Answer" (Log Transformation):

- Remember how our GDP data was heavily skewed, with some countries having extremely high GDPs and most having lower ones? To help our models learn more effectively, we apply a mathematical trick called a **logarithmic transformation** to the GDP values.
- Think of it like looking at a very wide mountain range from a distance: some peaks are much higher than others. A log transformation compresses these extreme differences, making the landscape appear more even, which allows the model to learn the general patterns better.

Dividing the Data for Learning and Testing (Train-Test Split):

Just like a student needs to study before taking a test, our model needs data to learn from. We split our prepared data into two main parts:

- **Training Data (80%)**: This is the larger portion that the model "studies" to learn the relationships between the clues and the answer.
- **Testing Data (20%)**: This is a completely separate portion that the model has never seen before. We use it to evaluate how well the model generalizes and makes predictions on new, unseen data, much like a final exam.

"Shrinking" Our Clues and Answers (Min-Max Scaling):

- Our economic indicators have very different scales (e.g., millions vs. billions). If we feed these raw numbers into the model, it might unfairly prioritize features with larger values.
- **Min-Max Scaling** is like giving all our clues and answers a uniform range (typically between 0 and 1). This ensures that every piece of information contributes fairly to the learning process, preventing any single feature from dominating due to its magnitude. It's like normalizing all the inputs so they speak the same numerical language.

By carefully performing these steps, we ensure our data is in the best possible shape for our machine learning models to learn and make accurate predictions about GDP.

Using the Basic Model:

(Random Forest Regression): The "Wisdom of Crowds" Approach

To predict Gross Domestic Product (GDP), we've employed a powerful machine learning technique called **Random Forest**. Imagine you need to make an important decision, so you ask a group of diverse experts for their individual opinions and then combine their answers to get a much more reliable final decision. That's essentially how a Random Forest works!

Here's a breakdown of how it operates:

1.Many "Decision Trees" (The Experts):

A Random Forest is an "ensemble" model, meaning it's made up of many individual decision-makers, called **Decision Trees**. Each tree is like a flowchart that asks a series of "yes" or "no" questions about our economic indicators (e.g., "Is agricultural output high?"). Based on the answers, it follows a path down the tree to arrive at a prediction for GDP.

2.Randomness in Building Trees (Diverse Experts):

To ensure our "experts" are diverse, a Random Forest introduces randomness in two key ways when building each tree:

- **Random Sample of Data:** Each tree is trained on a slightly different, random subset of our overall training data.
- **Random Subset of Features:** At each step of building a tree (each "question" in the flowchart), the tree only considers a random subset of our available economic indicators. This prevents any single indicator from dominating all the trees and encourages variety.

3.Averaging Predictions (Combining Wisdom)

- Once all the individual decision trees are built and trained, when we want to make a new GDP prediction, we feed the input economic indicators to every single tree in the forest.
- Each tree gives its own individual prediction. The Random Forest then takes all these individual predictions and **averages them together** to produce the final, more robust, and accurate GDP prediction.

Why is this so effective?

•**Reduces Overfitting:** By averaging many diverse trees, a Random Forest is less likely to simply memorize the training data (overfitting) and generally performs better on new, unseen data.

•**Handles Complex Relationships:** It can capture non-linear and intricate relationships between our economic indicators and GDP without requiring us to specify them beforehand.

•**Robustness:** It's more resilient to noise and outliers in the data because errors from one tree tend to be canceled out by the other trees.

Interpretation: Random Forest Regressor Performance

After training the **Random Forest Regressor** model on our feature-engineered data, we evaluated its performance using two key metrics:

•**R2 Score (Coefficient of Determination):** This metric indicates the proportion of the variance in the dependent variable (GDP) that is predictable from the independent variables (our economic indicators). An R2 score of 1 means the model perfectly predicts the target, while 0 means it explains none of the variance. Higher R2 is better.

•**MAE (Mean Absolute Error):** This is the average of the absolute differences between the predicted values and the actual values. It gives a direct measure of the average magnitude of the errors in the original scale of GDP. Lower MAE is better.
Our Random Forest Results:

•**R2 Score: ~0.9891** - An exceptionally high R2 score indicates that the Random Forest model explains approximately **98.9%** of the variance in Gross Domestic Product. This suggests the model is very effective at capturing the underlying patterns and relationships in the economic data.

•**MAE (Original Scale): ~832,404.19** - The Mean Absolute Error on the original scale of GDP is approximately **832,404.19**. This means, on average, our Random Forest model's predictions for GDP are off by roughly this amount. Given that GDP values can be in the millions or billions, this relatively low MAE signifies a **highly accurate prediction** in real-world terms, indicating the model's strong practical utility.

Predicting GDP with Neural Networks

What is Deep Learning?

Imagine teaching a child to recognize different animals. Instead of giving them a strict rulebook, you show them many pictures of cats, dogs, birds, etc., and tell them what each one is. Over time, the child learns to identify new animals on their own by picking up on subtle patterns. Deep learning works similarly! It's a type of machine learning that uses "neural networks" computer systems inspired by the human brain to learn from vast amounts of data. These networks can identify complex patterns and relationships that are difficult for humans or traditional computer programs to spot.
In our project, we're using deep learning to predict a country's Gross Domestic Product (GDP) by learning from historical agricultural economic indicators.

Building Our Neural Network: A "Brain" for GDP Prediction

Our neural network is structured like a series of interconnected layers, each with a specific role:

1.**Input Layer**: This is where we feed in our economic "clues" for a given country and year. These clues include current values of agricultural production, manufacturing, and past GDP values (like last year's GDP or the average GDP over the last three years).

2.**Hidden Layers (The "Thinkers")**: Our network has two of these. These layers are where the real "thinking" happens. They process the input clues, combine them in various ways, and identify intricate patterns. Each "neuron" in these layers acts like a tiny decision-maker, passing on its activated signal to the next layer. We've used a special activation function called Leaky Relu that helps these layers learn effectively without getting 'stuck'. We also added 'Dropout' layers, which randomly turn off some connections during training. This helps the network become more robust and prevents it from over-relying on any single piece of information, much like how a diverse team makes better decisions.

3.**Output Layer (The "Predictor")**: This is the final layer that takes all the processed information and generates our single, crucial "answer": the predicted Gross Domestic Product.
Think of it as a team of experts (the layers) collaborating to analyze all available information and arrive at the best possible prediction for GDP.

Training the Neural Network:

How the Neural Network Learns:

Just like a student needs to study and practice, our neural network needs to be trained on historical data to learn how to make accurate predictions.
Here's how we "teach" our model:

Setting the Learning Goal (loss=mean_squared_error): We tell the model what it's trying to minimize: the difference between its predictions and the actual GDP values. "Mean Squared Error" (MSE) means it tries to make these differences as small as possible, penalizing larger errors more heavily.

**The Coach (`optimizer=Adam (learning_rate=0.001))**: This is the algorithm that guides the learning process. The `Adam` optimizer is a popular and efficient coach that helps the network adjust its internal connections (weights) to reduce the error. The learning rate is like the coach's feedback intensity – a small rate means small adjustments, preventing big mistakes.

Study Sessions (epochs=100): An "epoch" is one complete pass through our entire training dataset. We run 100 such sessions, allowing the network to repeatedly see the data and refine its understanding.

Learning in Batches (batch_size=32): Instead of processing all data at once (which would be overwhelming), the network learns from small groups of 32 examples at a time. This makes the learning process more stable and efficient.

Practice Tests (validation_split=0.1): During training, we set aside a small portion (10%) of the training data as a "validation set." The model doesn't learn from this data, but we use it to check how well the model is generalizing to unseen examples during training. This helps us catch if the model is just memorizing the training data instead of truly learning.
After 100 epochs, the network has learned the complex relationships in the data and is ready to make predictions on new, unseen economic data.

Interpretation: Neural Network Performance

After training our **Neural Network** model, we evaluated its performance using the same key metrics as the Random Forest Regressor:

R2 Score (Coefficient of Determination): This metric tells us how well our model's predictions align with the actual data. An R2 of 1 means perfect prediction, while 0 means the model explains none of the variance in the target.

MAE (Mean Absolute Error): This represents the average absolute difference between the predicted GDP and the actual GDP, expressed in the original scale of GDP. A lower MAE indicates more accurate predictions.

Our Neural Network Results:

R2 Score: ~0.6378 - This R2 score indicates that the Neural Network model explains approximately **63.7%** of the variance in Gross Domestic Product. While this is a reasonably good score, it suggests that there is still a significant portion of the GDP's variability (about 36.3%) that the model is not capturing.

MAE (Original Scale): ~5,380,054.49 - The Mean Absolute Error on the original scale of GDP is approximately **5,380,054.49**. This means, on average, the Neural Network's predictions for GDP are off by roughly this amount. Compared to the Random Forest model's MAE, this value is considerably higher, indicating that the Neural Network's predictions are less accurate on average for this dataset.

Model Performance Comparison and Selection

We have trained and evaluated two distinct machine learning models for GDP prediction: The **Random Forest Regressor** and A **Neural Network**. Let's review their performance in the table below:

Model	R2 Score	MAE (Original Scale)
Random Forest Regressor	~0.9891	~832,404.19
Neural Network	~0.6378	~5,380,054.49

Key Observations:

Superior Accuracy: The **Random Forest Regressor** significantly outperformed the Neural Network, achieving a much higher R2 score (~0.9891 vs ~0.6378) and a substantially lower Mean Absolute Error (MAE) on the original GDP scale (~832k vs ~5.3M).

Robustness: The Random Forest's ensemble nature (combining multiple decision trees) made it more robust and effective at capturing the complex, non-linear relationships within our economic dataset, leading to more precise predictions.

Decision for Demo Model:

Given its demonstrably superior performance, the **Random Forest Regressor** has been selected as the model to power our interactive **Gradio Demo.** This choice ensures that the demo will showcase a highly accurate and reliable prediction tool for Gross Domestic Product, providing the most insightful experience for users.

GRADIO DEMO: GDP Prediction Tool

Purpose:

This interactive Gradio interface allows users to predict the Gross Domestic Product (GDP) based on various economic indicators. It leverages a trained **Random Forest Regressor** model that incorporates advanced feature engineering, including lagged values and rolling means, to provide more robust predictions. This tool is designed to offer insights into the relationship between agricultural and manufacturing output and a country's overall economic health (GDP).

How to Use:

1. **Adjust Input Values:** Modify the numerical values in the input fields according to the economic scenario you wish to analyze.
2. **Observe Predictions:** The interface will automatically update the "Predicted Gross Domestic Product" output as you change the input values.

Input Fields:

The model uses 11 input features to make its prediction. Each input represents a specific economic indicator or a derived feature from these indicators:

- Value Added (Agriculture): The current year's value added by the agriculture sector.
- Value Added (Agriculture, Forestry and Fishing): The current year's value added by the combined agriculture, forestry, and fishing sectors.
- Value Added (Manufacture of food and beverages): The current year's value added by the food and beverage manufacturing sector.
- Gross Domestic Product (Lag 1 Year): The GDP value from the previous year. This lagged feature helps the model understand economic trends and momentum.
- Value Added (Agriculture) (Lag 1 Year): The agriculture value-added from the previous year.
- Value Added (Agriculture, Forestry and Fishing) (Lag 1 Year): The agriculture, forestry, and fishing value-added from the previous year.
- Value Added (Manufacture of food and beverages) (Lag 1 Year): The food and beverage manufacturing value-added from the previous year.
- Gross Domestic Product (3-Year Rolling Mean): The average GDP over the past three years. Rolling means capture longer-term trends and smooth out short-term fluctuations.
- Value Added (Agriculture) (3-Year Rolling Mean): The 3-year rolling mean of agriculture value-added.
- Value Added (Agriculture, Forestry and Fishing) (3-Year Rolling Mean): The 3-year rolling mean of agriculture, forestry, and fishing value-added.
- Value Added (Manufacture of food and beverages) (3-Year Rolling Mean): The 3-year rolling mean of food and beverage manufacturing value-added.

Default Values: Each input field is pre-filled with the mean value of that feature from the training dataset. You can use these as a starting point or enter your own values.

Output:

Predicted Gross Domestic Product: This displays the model's prediction for the current year's GDP, based on the provided input economic indicators. The output is in the original scale (e.g., millions of USD) and represents the un-transformed value.

Model Information:

The prediction is powered by a **Random Forest Regressor** model. Random Forests are an ensemble learning method that builds multiple decision trees and merges their predictions to get a more accurate and stable forecast. This model was trained on historical economic data, including feature-engineered variables like 1-year lags and 3-year rolling means, and the target variable (GDP) was log-transformed and then scaled to improve model performance and handle data skewness.

Explaining the Gradio Input Constraints

1. User-Friendly Defaults (Mean Values):
Notice that when you first open the interface, all input fields are pre-filled with numbers. These are the **average (mean) values** for each economic indicator from our historical training dataset. This gives you a quick starting point to see how the model predicts GDP under 'typical' conditions."

2. Interactive Prediction:
To use the tool, simply change any of these numbers. For example, if you want to see the impact of higher agricultural output, increase the 'Value Added (Agriculture)' input. The model will instantly recalculate and display a new 'Predicted Gross Domestic Product'.

3. Guided Inputs (Min/Max Constraints):
A key improvement we've made is to add **minimum and maximum boundaries** for each input. These boundaries correspond to the lowest and highest values observed for that indicator in our actual training data."

Why is this important? It helps ensure that any values you enter are realistic and within the range of what the model has learned from. Entering values outside these historical ranges could lead to less reliable predictions, so these limits guide you towards meaningful inputs."

Instructions for using the Gradio GDP Prediction Interface:

Default Values (Mean Values):
When the interface first loads, all input fields are pre-filled with the average (mean) values for each economic indicator. These means are calculated directly from the 'X' (features) DataFrame, which represents the dataset used to train the model. This provides a typical starting point for predictions.

User Interaction:
To make a prediction, simply adjust the numerical values in any of the input fields. You can change one or multiple indicators. After modifying the inputs, click the 'Submit' or 'Predict' button to get an updated 'Predicted Gross Domestic Product' value.

Min/Max Constraints:
Each input field is constrained by a minimum and maximum value. These limits correspond to the lowest and highest values observed for that specific indicator in the 'X' training dataset.

Purpose of Constraints: These boundaries help guide users to provide realistic inputs that the model has been trained on. Inputting values outside these ranges might lead to less reliable predictions, as the model has not seen such data during its training.

By understanding these aspects, users can effectively interact with the interface to explore GDP predictions under various economic scenarios.

LINK: https://d1d3f4da858cdde965.gradio.live/

Conclusion:

Summary of Findings

•**Predictive Power:** Our model successfully translated complex agricultural indicators into reliable GDP forecasts, achieving high accuracy through neural network optimization.

•**The Agriculture-Wealth Link:** The data confirms that for many developing economies, agricultural stability is not just a sector, it is the primary engine for national economic growth.

•This project demonstrates that by monitoring agricultural sector in real-time, policymakers can anticipate economic shifts before they appear in traditional financial reports.

To wrap things up, we didn't just build a model; we proved that we can use machine learning to turn specialized sector data into a global economic forecast.

THE END

THANK YOU